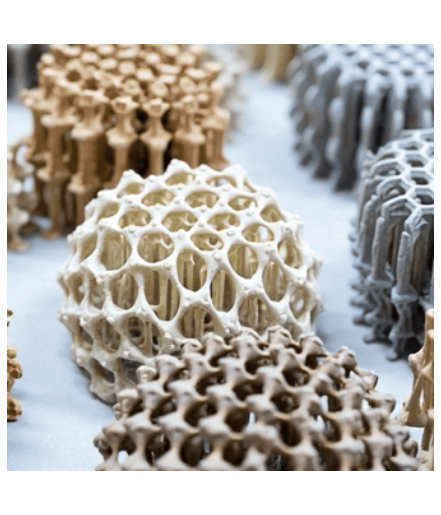
> Technical skill is the mastery of complexity, while creativity is the mastery of simplicity.
>
> — Christopher Zeeman

Engineering Capabilities Hands-on Expertise Technical Standards & Quality Assurance Important Links

ENGINEERING CAPABILITIES

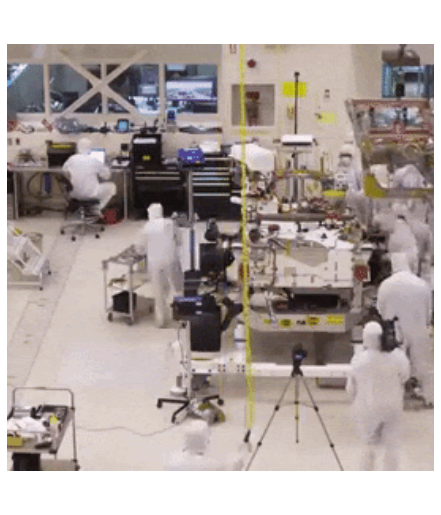


Metamaterials/ Lattice Structures

- Architected Structures
- Porous Structures
- Lattice Structures (including TPMS types such as Primitive, Gyroid, and Diamond)

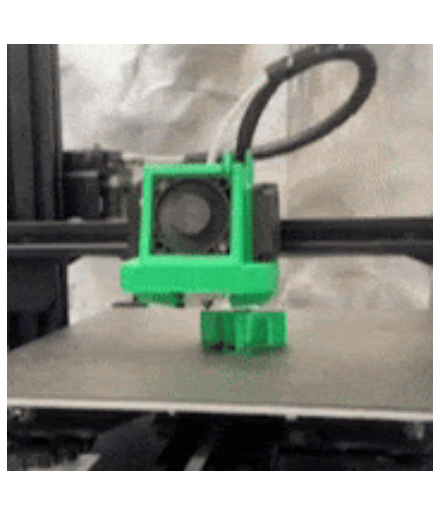
Mechanical Testing

- Compression Testing
- Tensile Testing
- Bending Testing
- Fatigue Testing
  (Conducted using Instron, MTS, and Controls universal testing machines)



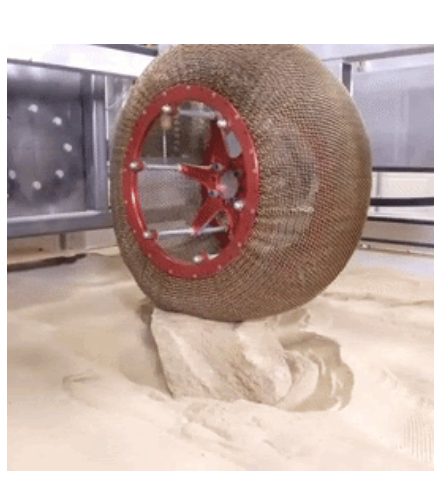
Experimental Research

- 3D Printing of polymer, ceramic, and metal samples
- Thermal Testing
- Mechanical Testing
- Heat Treatment
- Data Analysis
- Research Paper Publications



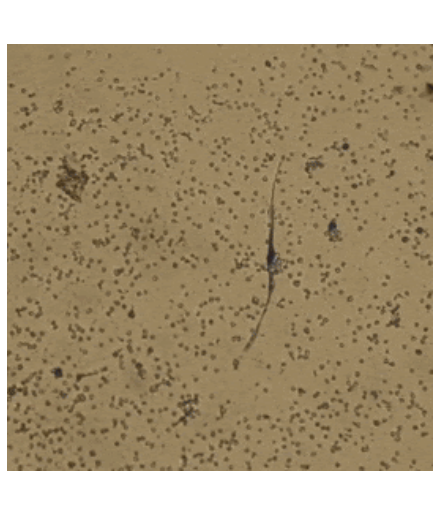
Additive Manufacturing

- Laser Powder Bed Fusion
- 3D Printing
- Metal Printing
- Polymer Printing



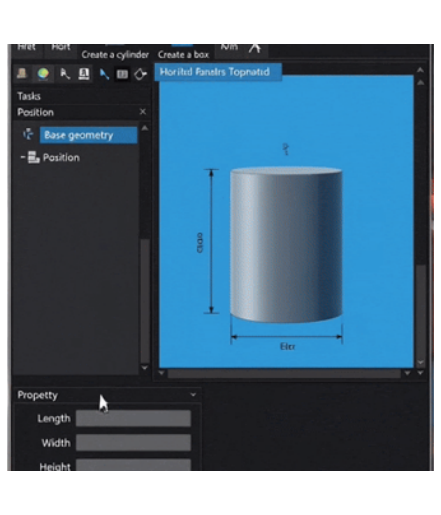
Shape Memory Alloys

- Shape Memory Alloys: Cu-Al-Mn, Cu-Al-Ni, NiTi
- Fabrication Methods: Vacuum Induction Melting, Sand Casting, Additive Manufacturing



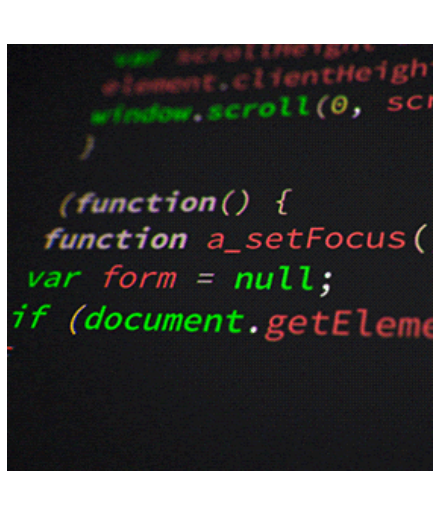
Materials Characterization

- Metallography
- Optical Microscopy
- Optical Emission Spectroscopy (OES)
- Scanning Electron Microscopy (SEM)
- X-ray Diffraction (XRD)
- Differential Scanning Calorimetry (DSC)
- Tensile Testing
- Compression Testing
- Fatigue Testing



FreeCAD

- Computer-Aided Design (CAD)
- Computer-Aided Manufacturing (CAM)
- Computer-Aided Engineering (CAE)

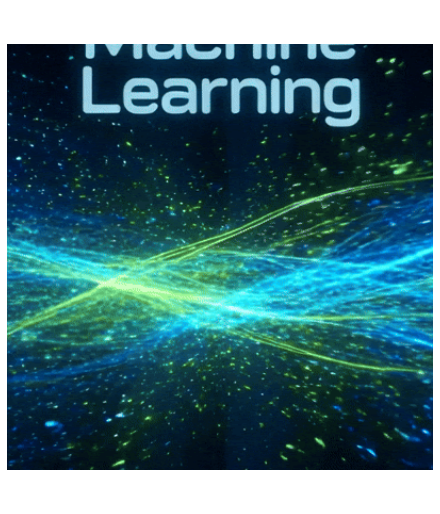


Python/MATLAB

- Data Analysis using Programming Languages
- Data Visualization using Programming Languages

Data Science

- Languages: Python, SQL, Shell Scripting
- Data Analysis: Pandas, NumPy, SciPy, BeautifulSoup
- Visualization: Matplotlib, Seaborn, Plotly, Dash, Folium
- Machine Learning & Tools: Scikit-learn, Jupyter, Anaconda
- Software Development: Git, GitHub, VS Code, Spyder, LaTeX
- OS: Linux



Machine Learning

- Supervised Learning
- Neural Networks & AI
- Advanced Algorithms
- Unsupervised & Reinforcement Learning

HANDS-ON EXPERTISE

Stratasys J850

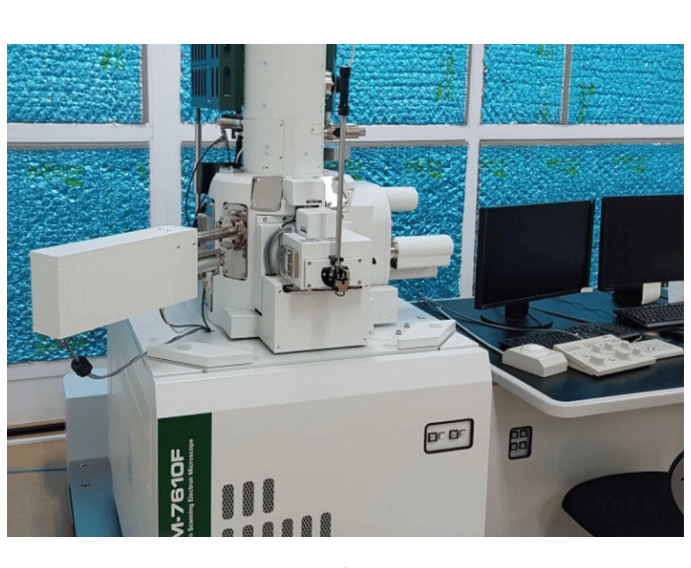
3D Printing and Slicing

- Formlabs Form3 Printer
- Ultimaker S1 Printer
- Stratasys F370 Printer
- Stratasys J850 Printer
- Raise3D Pro3 Plus HS Printer
- Raise3D Forge1 Printer
- TinkerCAD
- FreeCAD
- Materialise Magics
- Cura Slicer
- IdeaMaker Slicer
- Grabcad Print Slicer

Controls Simplex 350

Mechanical Processing and Testing

- Instron 5969 Static UTM 50 kN loadcell
- Instron 8872 fatigue loading machine 25 kN loadcell
- MTS 100 kN loadcell
- Controls Simplex 350
- Controls Automax Compression Machine
- Two roll Rolling Machine
- Vacuum Induction Melting Machine



JEOL 7610F SEM

Materials Characterization

- JEOL 7610F SEM
- FEI Nova NanoSEM
- FEI Quanta 3D FIB SEM
- Bruker D2 Phasor XRD
- Rigaku MiniFlex II XRD
- Xpert HighScore Plus Software
- Leica Optical Microscope
- Setaram Instruments DSC
- Mettler Toledo DSC 1
- Bruker Q4 Tasman

Struers Grinder and Polisher

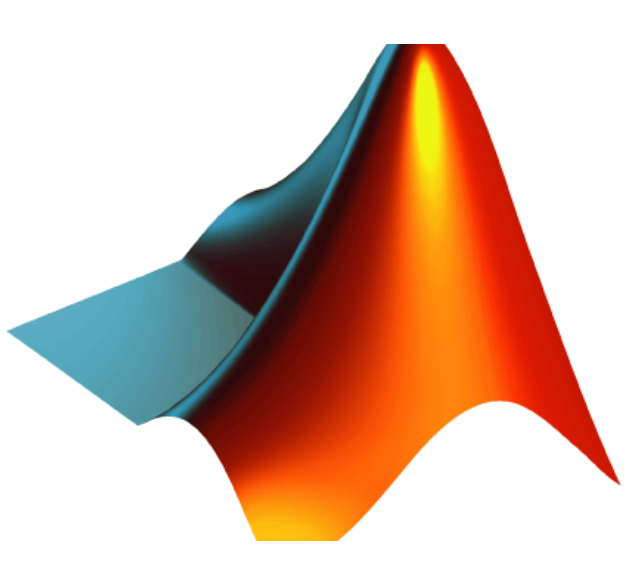
Metallography

- Struers Grinder and Polisher
- Buehler Grinder and Polisher
- Chemical Etching
- Heat Treatment (Carbolite Furnaces)

Shell Scripting

High Performance Computing

- Python
- Linux
- Shell Scripting
- FreeCAD
- ANSYS
- MATLAB
- Originlab



MATLAB

Image Processing

- Python
- OpenCV
- Scikit-image
- ImageJ/Fiji
- MATLAB

Numpy

Data Science and Machine Learning

- JupyterLab and Notebook
- Anaconda
- Python
- Pandas
- Numpy
- Matplotlib
- Plotly
- Seaborn
- Scipy
- Scikit-learn
- TensorFlow
- Folium
- SQLite3
- LaTeX

TECHNICAL STANDARDS & QA

Additive Manufacturing (AM) Frameworks

- **ISO/ASTM 52900:** Expert knowledge in general principles and terminology for additive manufacturing processes.
- **ISO/ASTM 52904:** Proficient in metal Powder Bed Fusion (LPBF) process characteristics and quality control for **EOS M400** operations.
- **ISO/ASTM 52903-1:** Standardized material extrusion-based manufacturing for high-performance polymers like **PPA/CF**.

Mechanical Testing & Validation

- **Metals:** Tensile and compression testing of metallic dogbones and cylinders according to **ASTM E8/E8M** and **ASTM E9**.
- **Polymers & Composites:** Characterization of rigid plastics and 3D-printed TPMS lattices following **ASTM D638** (tensile) and **ASTM D695** (compression).
- **Fatigue:** Strain-controlled and cyclic fatigue testing of NiTi components adhering to **ASTM E606** and **ISO 12106**.
- **Flexural:** Evaluated structural integrity of cast aluminum and 3D-printed beams using 4-point bending protocols per **ASTM E290**.

Material Characterization (Thermal & Structural)

- **Phase Transformation (SMAs):** Analysis of NiTi and Cu-based shape memory alloys via Differential Scanning Calorimetry (DSC) in compliance with **ASTM F2004** and **ASTM F2082**.
- **Microstructural Analysis:** Advanced microscopy and diffraction techniques (SEM/XRD) focused on phase identification and grain boundary analysis.

IMPORTANT LINKS

- Timeline of scientific discoveries
- Recommended books on 3D printing
- Materials characterization tools training
- FreeCAD tutorials for computer aided designing
- FEM (finite element method/analysis) tutorials
- Learn to code HTML and CSS
- Learn VIM
- Dr Chuck online
- Building a modern computer from first principles
- Circuit making
- Python libaries for image processing
- Python libraries for scientific computing

Academic transcript by Credly

Learn more about me

Explore the Data Science projects

Shahadat Hussain, PhD

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Exploring advanced materials for innovative applications

Key Profiles

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- GitHub
- Grokipedia
- Institutional Page

Navigation

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3D Printing

- Raise3D Pro3 Plus HS
- Raise3D Pro2 Plus
- Raise3D Forge1
- Ultimaker S1
- Stratasys F370
- Stratasys J850
- Formlabs Form3
- Bambu Lab Carbon X1

Design & Simulation

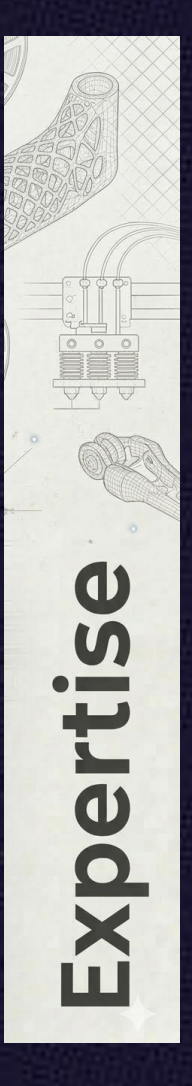
- FreeCAD
- Tinkercad
- MSLattice
- SolidWorks
- Materialise Magics
- IdeaMaker
- Cura
- Grabcad Print
- PreForm
- Bambu Studio

Data Science

- MATLAB
- Python
- Pandas
- Numpy
- Matplotlib
- Plotly
- Seaborn
- Scipy
- Scikit-learn
- SQLite
- Jupyter Notebook
- Linux
- Shell Scripting
- LaTeX

Mechanical Testing

- Instron 5969 (Bluehill Universal)
- Instron 8872 (WaveMatrix3)
- MTS 100 kN (TestSuite TW)
- Controls Simplex 350 & Automax (Multitest)
- KB 250 BVRZ (HardWin XL)



AVAILABILITY
Open for Collaborative Research and Industrial Consultancy.

Professional Networks & Links

Active across research networks, design & development platforms.

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Business Card

Micro-Credentials

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